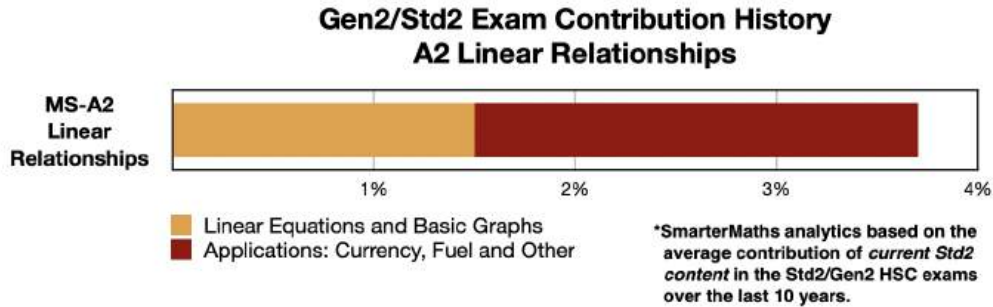


STD 1: Algebra (Std 1), A2 Linear Relationships (Y11)

Linear Equations and Basic Graphs (Std 1)

Teacher: Denis Vlismas

Exam Equivalent Time: 45 minutes (based on HSC allocation of 1.5 minutes approx. per mark)



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-A2 Linear Relationships has accounted for 5% and 4% of the 2020 and 2019 Std1 exams respectively.
- MS-A2 Linear Relationships is split into the two sub-topics listed in the graph above. Note that estimated exam contributions in the graph are taken from Std2/Gen2 past exams due to the very limited Std1 exam history. They should only be used for estimating the "relative" importance of each sub-topic.
- This analysis looks at *Linear Equations and Basic Graphs*.

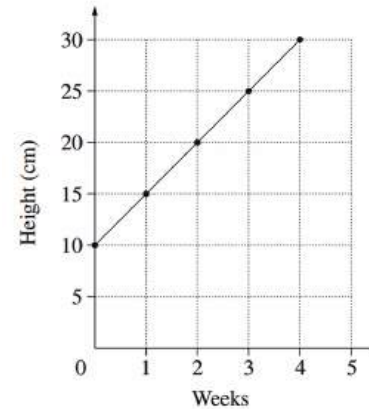
ANALYSIS - What to Expect and Common pitfalls

- Linear Equations and Basic Graphs* is a minor area that covers "gateway" knowledge which is required to solve harder practical problems found later in the course.
- Although not examined to date in Std1 exams, past Gen2/Std2 papers tested a basic conceptual understanding via multiple choice in 4 consecutive years up until 2017, but not since.
- The most common question types ask students to identify the graphs of simple linear equations or find gradients from simple graphs.
- It is very surprising how much trouble Std2 students have historically had with these questions (sub-50% mean marks half the time).

Questions

1. Algebra, STD2 A2 2004 HSC 2 MC

Susan drew a graph of the height of a plant.



What is the gradient of the line?

- (A) 1
- (B) 5
- (C) 7.5
- (D) 10

2. Algebra, STD2 A2 SM-Bank 1 MC

What is the slope of the line with equation $2x - 4y + 3 = 0$?

- (A) -2
- (B) $-\frac{1}{2}$
- (C) $\frac{1}{2}$
- (D) 2

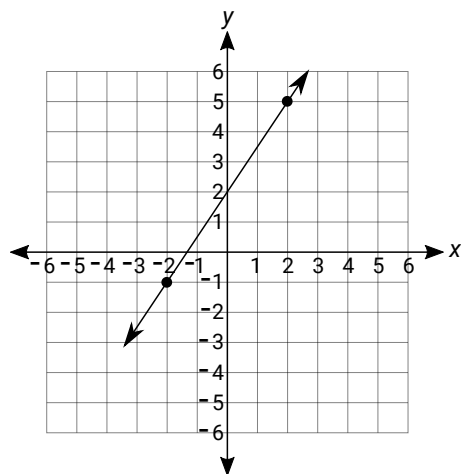
3. Algebra, STD2 A2 SM-Bank 2 MC

What is the gradient of the line $2x + 3y + 4 = 0$?

- (A) $-\frac{2}{3}$
 (B) $\frac{2}{3}$
 (C) $-\frac{3}{2}$
 (D) $\frac{3}{2}$

4. Algebra, STD2 A2 SM-Bank 11 MC

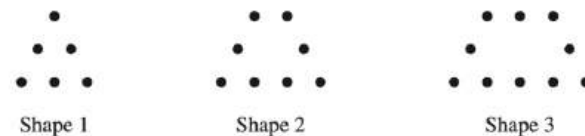
What is the equation for the line shown?



- A. $y = -\frac{3}{2}x + 2$
 B. $y = \frac{3}{2}x + 2$
 C. $y = -\frac{2}{3}x + 2$
 D. $y = \frac{2}{3}x + 2$

5. Algebra, STD2 A2 2012 HSC 8 MC

Dots were used to create a pattern. The first three shapes in the pattern are shown.



The number of dots used in each shape is recorded in the table.

Shape (S)	1	2	3
Number of dots (N)	6	8	10

How many dots would be required for Shape 156?

- (A) 316
 (B) 520
 (C) 624
 (D) 936

6. Algebra, STD2 A2 2006 HSC 7 MC

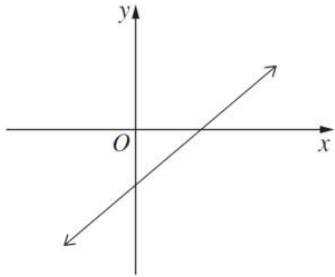
Which equation represents the relationship between x and y in this table?

x	0	2	4	6	8
y	1	2	3	4	5

- (A) $y = 2x + 1$
 (B) $y = 2x - 2$
 (C) $y = \frac{x}{2} - 2$
 (D) $y = \frac{x}{2} + 1$

7. Algebra, STD2 A2 2016 HSC 14 MC

The graph shows a line which has an equation in the form $y = mx + b$.

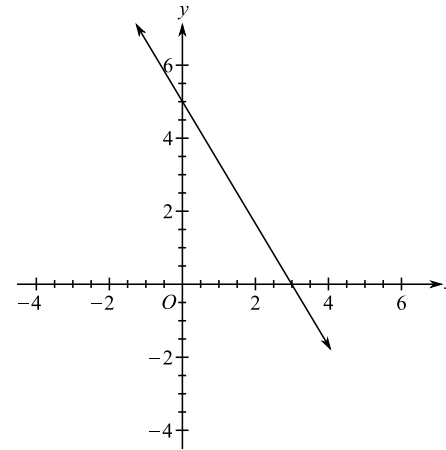


Which of the following statements is true?

- (A) m is positive and b is negative
 - (B) m is negative and b is positive
 - (C) m and b are both positive
 - (D) m and b are both negative
-

8. Algebra, STD2 A2 SM-Bank 5 MC

The equation of the line drawn in the diagram below is:



- (A) $5y = 3x$
 - (B) $y = \frac{5}{3}x + 5$
 - (C) $y = \frac{3}{5}x + 5$
 - (D) $y = -\frac{5}{3}x + 5$
-

9. Algebra, STD2 A2 2020 HSC 6 MC

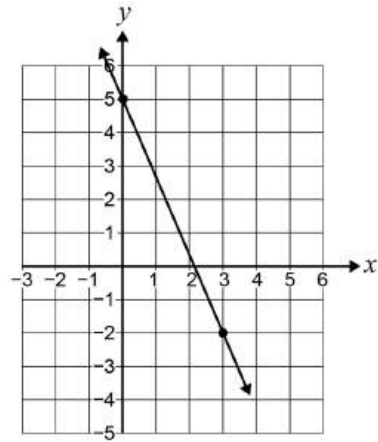
Suppose $y = -1 - 2x$.

When the value of x increases by 5, the value of y decreases by

- A. 1.
 - B. 2.
 - C. 5.
 - D. 10.
-

10. Algebra, STD2 A2 SM-Bank 10 MC

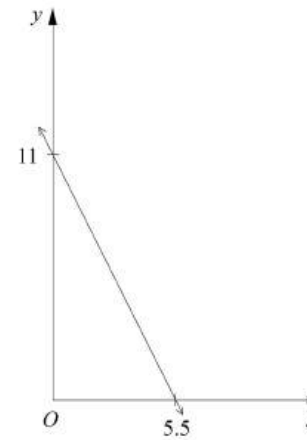
Leo drew a straight line through the points (0, 5) and (3, -2) as shown in the diagram below.



What is the gradient of the line that Leo drew?

- A. $\frac{7}{3}$
- B. $\frac{3}{7}$
- C. $-\frac{3}{7}$
- D. $-\frac{7}{3}$

11. Algebra, STD2 A2 SM-Bank 12 MC



Which of these equations represents the line in the graph?

- A. $y = 11 - 2x$
- B. $y = 11 + 5.5x$
- C. $y = 11 - 5.5x$
- D. $y = 11 + 2x$

12. Algebra, STD2 A2 SM-Bank 6 MC

Renee went bike riding on a holiday.

The hiring charges are listed in the table below:

Hours hired (h)	1	2	3	4	5
cost (C)	18	24	30	36	42

Which linear equation shows the relationship between C and h ?

- A. $C = 12 + 6h$
- B. $C = 6 + 12h$
- C. $C = 18 + 12h$
- D. $C = 12 + 18h$

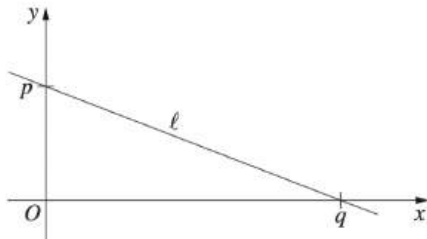
13. Algebra, STD2 A2 2009 HSC 14 MC

If $A = 6x + 10$, and x is increased by 2, what will be the corresponding increase in A ?

- (A) $2x$
- (B) $6x$
- (C) 2
- (D) 12

14. Algebra, STD2 A2 2012 HSC 5 MC

The line l has intercepts p and q , where p and q are positive integers.



What is the gradient of line l ?

- (A) $-\frac{p}{q}$
- (B) $-\frac{q}{p}$
- (C) $\frac{p}{q}$
- (D) $\frac{q}{p}$

15. Algebra, STD2 A2 2005 HSC 17 MC

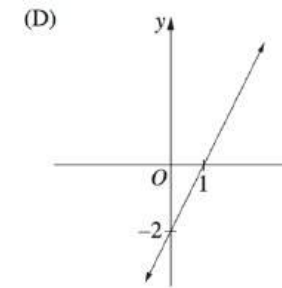
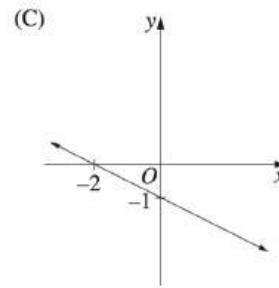
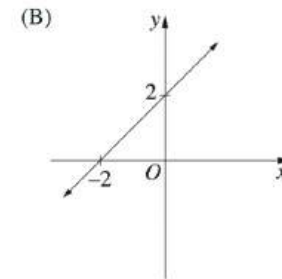
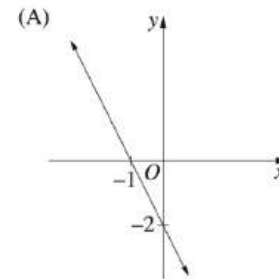
The total cost, $\$C$, of a school excursion is given by $C = 2n + 5$, where n is the number of students.

If three extra students go on the excursion, by how much does the total cost increase?

- (A) \$6
- (B) \$11
- (C) \$15
- (D) \$16

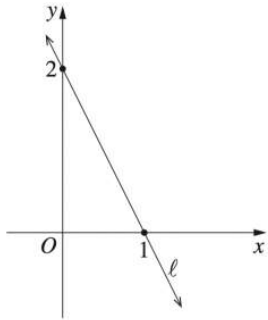
16. Algebra, STD2 A2 2014 HSC 7 MC

Which of the following is the graph of $y = 2x - 2$?



17. Algebra, STD2 A2 2015 HSC 13 MC

What is the equation of the line ℓ ?



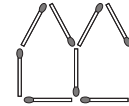
- (A) $y = -2x + 2$
 - (B) $y = 2x + 2$
 - (C) $y = -\frac{x}{2} + 2$
 - (D) $y = \frac{x}{2} + 2$
-

18. Algebra, STD2 A2 2017 HSC 20 MC

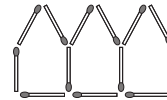
A pentagon is created using matches.



By adding more matches, a row of two pentagons is formed.



Continuing to add matches, a row of three pentagons can be formed.

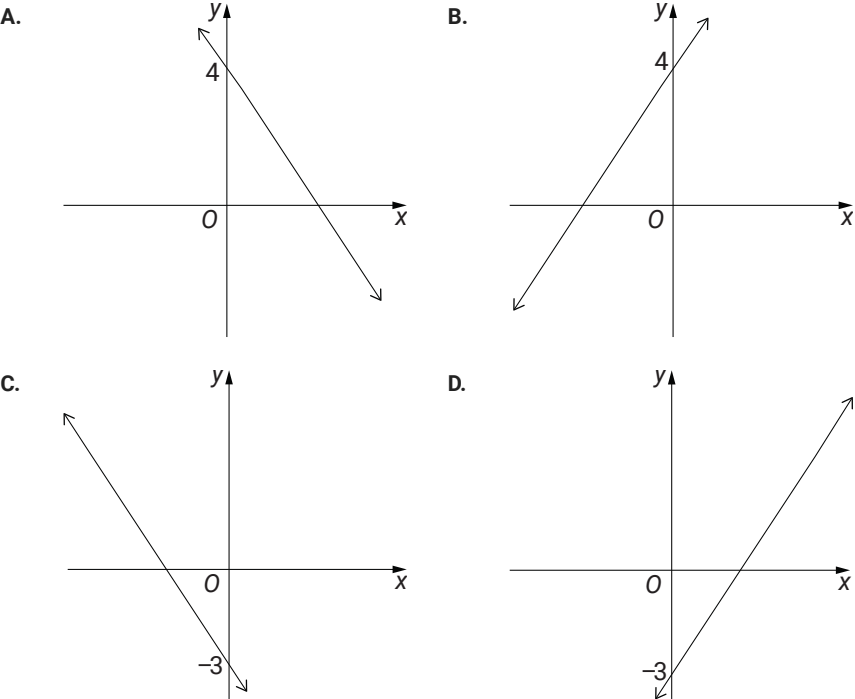


Continuing this pattern, what is the maximum number of complete pentagons that can be formed if 100 matches in total are available?

- A. 25
 - B. 24
 - C. 21
 - D. 20
-

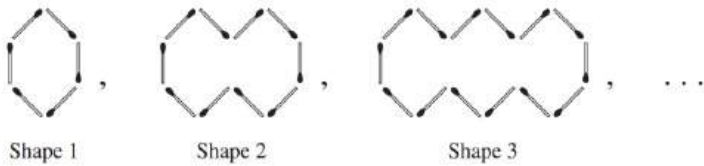
19. Algebra, STD1 A2 2021 HSC 5 MC

Which of the following best represents the graph of $y = -3x + 4$?



20. Algebra, STD2 A2 2007 HSC 18 MC

Chris started to make this pattern of shapes using matchsticks.



If the pattern of shapes is continued, which shape would use exactly 486 matchsticks?

- (A) Shape 96
- (B) Shape 97
- (C) Shape 121
- (D) Shape 122

21. Algebra, STD2 A2 SM-Bank 1

Sketch the graph of $y - 2x = 3$, showing the intercepts on both axes. (2 marks)

22. Algebra, STD2 A2 SM-Bank 20

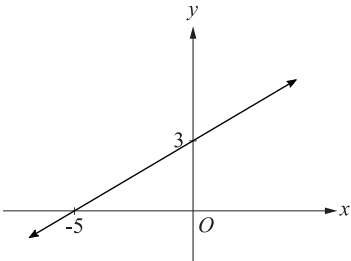
Peter uses matchsticks to make a pattern of shapes, as shown in the table below.

Shape number (N)	1	2	3
			
Number of Sticks (S)	6	11	16

How many sticks (S) will be needed to make Shape Number (N) 13?

23. Algebra, STD2 A2 SM-Bank 5

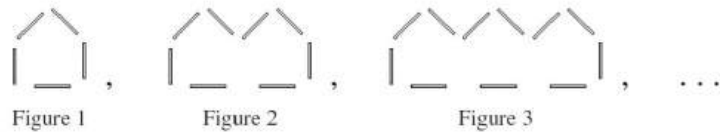
The diagram shows the graph of a line.



What is the equation of this line? (2 marks)

24. Algebra, STD2 A2 2011 HSC 23b

Sticks were used to create the following pattern.



The number of sticks used is recorded in the table.

Figure (F)	1	2	3
Number of sticks (N)	5	8	11

- Draw Figure 4 of this pattern. (1 mark)
- How many sticks would be required for Figure 100? (1 mark)
- Is it possible to create a figure in this pattern using exactly 543 sticks?
Show suitable calculations to support your answer. (2 marks)

Worked Solutions

- Algebra, STD2 A2 2004 HSC 2 MC

2 points on graph (0, 10), (1, 15)

$$\begin{aligned}\text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{15 - 10}{1 - 0} \\ &= 5\end{aligned}$$

$\Rightarrow B$

- Algebra, STD2 A2 SM-Bank 1 MC

$$2x - 4y + 3 = 0$$

$$4y = 2x + 3$$

$$y = \frac{1}{2}x + \frac{3}{4}$$

$$\therefore \text{Slope} = \frac{1}{2}$$

$\Rightarrow C$

- Algebra, STD2 A2 SM-Bank 2 MC

$$2x + 3y + 4 = 0$$

$$3y = -2x - 4$$

$$y = -\frac{2}{3}x - \frac{4}{3}$$

$$\therefore \text{Gradient} = -\frac{2}{3}$$

$\Rightarrow A$

- Algebra, STD2 A2 SM-Bank 11 MC

Graph cuts y -axis at 2

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{3}{2}$$

$$\therefore y = \frac{3}{2}x + 2$$

$\Rightarrow B$

5. Algebra, STD2 A2 2012 HSC 8 MC

Linear relationship where

$$N = 4 + (2 \times S)$$

When $S = 156$

$$N = 4 + (2 \times 156)$$

$$= 316$$

$$\Rightarrow A$$

6. Algebra, STD2 A2 2006 HSC 7 MC

By elimination (using the table)

$(0, 1)$ must satisfy

$\therefore$ NOT B or C

$(2, 2)$ must satisfy

$\therefore$ NOT A

$$\Rightarrow D$$

7. Algebra, STD2 A2 2016 HSC 14 MC

$$\Rightarrow A$$

8. Algebra, STD2 A2 SM-Bank 5 MC

y -intercept = $+5$

$$\text{Gradient} = \frac{\text{rise}}{\text{run}}$$

$$= -\frac{5}{3}$$

$$\therefore \text{Equation is: } y = -\frac{5}{3}x + 5$$

$$\Rightarrow D$$

9. Algebra, STD2 A2 2020 HSC 6 MC

Strategy 1

If x increases by 5

$$\Rightarrow y \text{ decreases by } 2x = 2 \times 5 = 10$$

Strategy 2

Test 2 values:

$$\text{If } x = 0, y = -1$$

$$\text{If } x = 5, y = -1 - 10 = -11$$

$\therefore y$ decreases by 10.

10. Algebra, STD2 A2 SM-Bank 10 MC

Line passes through $(0, 5)$ and $(3, -2)$

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5 - (-2)}{0 - 3}$$

$$= -\frac{7}{3}$$

11. Algebra, STD2 A2 SM-Bank 12 MC

Graph passes through $(0, 11)$ and $(5.5, 0)$

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{11 - 0}{0 - 5.5}$$

$$= -2$$

$$\therefore \text{Equation is: } y = 11 - 2x$$

$$\Rightarrow A$$

12. Algebra, STD2 A2 SM-Bank 6 MC

Consider Option 1:

$$12 + (6 \times 1) = 12 + 6 = 18$$

$$12 + (6 \times 2) = 12 + 12 = 24$$

$$12 + (6 \times 3) = 12 + 18 = 30 \quad \text{etc ...}$$

$\therefore$ The linear equation is: $C = 12 + 6h$

$\Rightarrow A$

13. Algebra, STD2 A2 2009 HSC 14 MC

$$A = 6x + 10$$

If x increases by 2

$$A \text{ increases by } 6 \times 2 = 12$$

$\Rightarrow D$

♦ Mean mark 50%.
STRATEGY: Substituting real numbers into the equation can work well in these type of questions. eg. If $x = 0$, $A = 10$ and when $x = 2$, $A = 22$

14. Algebra, STD2 A2 2012 HSC 5 MC

$$\text{Gradient} = \frac{\text{rise}}{\text{run}}$$

$$= -\frac{p}{q}$$

$\Rightarrow A$

♦ Mean mark 45%

15. Algebra, STD2 A2 2005 HSC 17 MC

$$C = 2n + 5$$

If n increases to $n + 3$

$$C = 2(n + 3) + 5$$

$$= 2n + 6 + 5$$

$$= 2n + 11$$

$\therefore$ Total cost increases by \$6

$\Rightarrow A$

16. Algebra, STD2 A2 2014 HSC 7 MC

$$y = 2x - 2$$

By elimination

It has a y intercept of -2

$\Rightarrow$ Cannot be B or C

$(-1, 0)$ from A doesn't satisfy equation

but $(1, 0)$ from D does

$\Rightarrow D$

♦ Mean mark 46%

17. Algebra, STD2 A2 2015 HSC 13 MC

l passes through $(0, 2)$ and $(1, 0)$

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{0 - 2}{1 - 0}$$

$$= -2$$

y intercept = 2

$$\therefore y = -2x + 2$$

$\Rightarrow A$

♦ Mean mark 48%

18. Algebra, STD2 A2 2017 HSC 20 MC

1 pentagon: 5 matches

2 pentagons: $5 + 4 = 9$

3 pentagons: $5 + 4 \times 2 = 13$

$\vdots$

n pentagons: $5 + 4(n - 1)$

$$5 + 4(n - 1) = 100$$

$$4n - 4 = 95$$

$$4n = 99$$

$$n = 24.75$$

$\therefore$ Complete pentagons possible = 24

$\Rightarrow B$

19. Algebra, STD1 A2 2021 HSC 5 MC

$$y = -3x + 4$$

$$y\text{-intercept} = 4$$

Gradient is negative $\rightarrow$ slopes from upper left to lower right

$$\Rightarrow A$$

◆◆ Mean mark 16%.

20. Algebra, STD2 A2 2007 HSC 18 MC

Shape (S)	1	2	3
Matches (M)	6	10	14

Equation rule:

$$M = 4S + 2$$

Find x when $M = 486$:

$$486 = 4S + 2$$

$$4S = 484$$

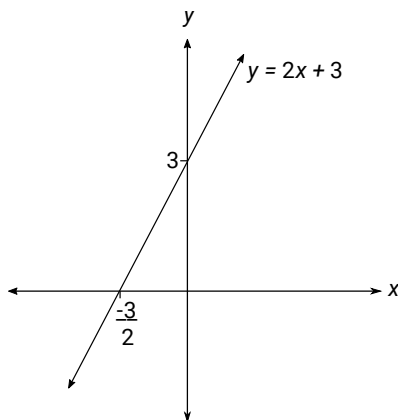
$$S = 121$$

$\therefore$ The 121st shape uses 486 matchsticks.

$$\Rightarrow C$$

21. Algebra, STD2 A2 SM-Bank 1

$$y - 2x = 3 \Rightarrow y = 2x + 3$$



22. Algebra, STD2 A2 SM-Bank 20

$$\text{Shape 1: } S = 5 \times 1 + 1 = 6$$

$$\text{Shape 2: } S = 5 \times 2 + 1 = 11$$

$\vdots$

$$\text{Shape } N: S = 5N + 1$$

$\therefore$ Sticks required when $N = 13$

$$= 13 \times 5 + 1$$

$$= 66$$

23. Algebra, STD2 A2 SM-Bank 5

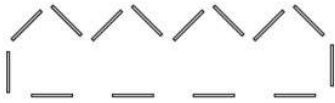
$$y\text{-intercept} = 3$$

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{3}{5}$$

$$\therefore \text{Equation: } y = \frac{3}{5}x + 3$$

24. Algebra, STD2 A2 2011 HSC 23b

i. Figure 4 is shown below:



ii. Since $N = 2 + 3F$

If $F = 100$,

$$\begin{aligned} N &= 2 + (3 \times 100) \\ &= 302 \end{aligned}$$

iii. $543 = 2 + 3F$

$$3F = 541$$

$$F = 180\frac{1}{3}$$

Since F is not a whole number, 543 sticks
will not create a figure.

♦ Mean mark 48%.
MARKER'S COMMENT: Students
should attempt to find a "rule" in
such questions, and use this
formula to solve the question, as
per the Worked Solution.